GMI German-Malaysian Institute

DIPLOMA PROGRAMME

ASSIGNMENT & REPORT

Academic Period: July 2026

Code	**NTC 1052**	**Course Name**	**NETWORK MANAGEMENT & MONITORING SYSTEMS**
Title	**FCAPS, SNMP & NETWORK DESIGN FRAMEWORKS**	**Examiner**	**Ms. NUR DAYANA AHMAD DAUD**
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No	Assessment Criteria	CLO	PO	Level of Diff	Full Marks	Score	Marks	Comment
1.	FCAPS & Design Analysis	1	3		2			
2.	SNMP Understanding	1	3		2			
3.	Tools Investigation	1	3		2			
4.	Report Formatting	3	9		2			
5.	Defence Ability	3	9		2			
				Total	**10**			
				Total Marks				
				Signature				

OVERVIEW

In modern enterprise environments, network engineers do not just define concepts—they architect scalable topologies and implement active monitoring frameworks to maintain high availability and security. This assignment requires you to analyze a multi-site enterprise scenario, map FCAPS management functions to active protocols, evaluate SNMP security vulnerabilities across versions, and justify an enterprise network architecture.

By completing this assignment, you will demonstrate the ability to:

1. Map real-world network operational challenges to specific **FCAPS management domains** and propose concrete tool/protocol solutions.
2. Evaluate **SNMP operational mechanisms** (Traps vs. Polling, MIB/OID parsing) and perform a security audit comparing SNMPv1, v2c, and v3.
3. Architect and justify an enterprise network using **Hierarchical, Centralized, or Distributed** design models under specific technical constraints.

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Part 1: FCAPS Integration & Enterprise Network Design

Task 1.1: Applied FCAPS Scenario Mapping

A company is facing several network issues: line outages are undetected, bandwidth is slow during peak hours, unrecorded changes are made to switch settings, unknown devices join the network, and the IT team cannot track network usage by department.

1. Briefly explain each of the **5 FCAPS functions**:
    - **F**ault Management
    - **C**onfiguration Management
    - **A**ccounting Management
    - **P**erformance Management
    - **S**ecurity Management
2. For each function above, identify **one network problem** from the company's situation and suggest a practical solution or tool (e.g., Ping/Syslog, SSH passwords, Wireshark, SNMP monitoring).

Task 1.2: Network Design Comparison

Compare three network design models: **Hierarchical (3-Tier: Core, Distribution, Access)**, **Centralized**, and **Distributed**.

1. Construct a comparison table covering:
    - **Ease of Expansion (Scalability)**
    - **Reliability (What happens if a main router/server breaks?)**
    - **Cost to set up and maintain**
2. **Scenario:** A growing college campus needs to connect 3 separate buildings to a main server room. Which network design model would you recommend and why? (Provide 2 key reasons).

Part 2: Advanced SNMP Protocol & Security Investigation

Task 2.1: SNMP Fundamentals

1. Explain the difference between **SNMP Polling** (manager asking a device) and **SNMP Traps** (device alerting the manager). Give a practical example of when a Trap would be triggered.

2. Explain why **SNMPv3** is much safer to use in a company network compared to **SNMPv1** and **SNMPv2c**.

Task 2.2: Comparative Study of 3 SNMP Implementations

1. Choose **three (3)** SNMP-supported tools or hardware agents (e.g., Cisco IOS SNMP, Net-SNMP on Linux, PRTG Network Monitor, SolarWinds, or Zabbix).

   For **each** chosen tool, answer the following in clear bullet points:

   - **SNMP Versions Supported:** (v1, v2c, or v3?)
   - **Main Feature:** How does the tool collect data or display alerts to the network admin?
   - **Pros & Cons:** Name 1 key advantage and 1 key limitation of the tool.

Rubric for Assignment

NO.	CRITERIA	APPROACHES EXPECTATIONS [1]	MEETS EXPECTATIONS [2]	EXCELLENT [3]	MARKS
ASSIGNMENT					
1	**FCAPS & Topology Analysis**	Basic definitions incomplete or missing scenario answers.	Explains FCAPS and compares network designs correctly.	Clear explanations with accurate scenario solutions and practical design choices.	/3
2	**SNMP Understanding**	Confuses Polling and Traps; does not explain SNMPv3 security benefits clearly.	Correctly explains Polling, Traps, and basic SNMPv3 security features.	Clear distinction between Polling/Traps with well-explained SNMPv3 security features.	/3
3	**Tools Investigation**	Examines fewer than 3 tools or missing key details (versions/pros/cons).	Examines 3 tools with standard descriptions of features and pros/cons.	Thorough investigation of 3 tools with well-organized comparisons.	/3
4	**Report Formatting**	Poor organization, missing required sections or submitted in incorrect format.	Clean layout in PDF, includes required sections and basic references.	Excellent structure, clear tables/diagrams, neatly formatted PDF with references.	/3
5	**Defence ability/ presentation**	Not able to answer questions, not prepared and no confidence at all	Able to answer questions but with little preparation and confidence	Able to all answer questions very well and confidently. Very well prepared	/3
	TOTAL				/15

Part 1: FCAPS Integration & Enterprise Network Design

Task 1.1: Applied FCAPS Scenario Mapping

1. FCAPS Function Definitions

1. Fault Management - Network faults happen, so catching them before they can cause serious issues is critical. The ability to identify, isolate, log, and correct them is a necessary component to every network. Say a switch just drops offline out of nowhere, a good monitoring system picks that up fast and gets IT staff on it before users even notice.

2. Configuration Management - Every switch and router has a bunch of settings that need tracking, IP addresses, VLANs, configs, all of it. This is what keeps that organized. It also keeps a history of changes, which honestly saves a lot of headaches when something breaks and nobody remembers what got changed last week.

3. Accounting Management - This one's really just about usage tracking. Who's using how much bandwidth, which devices, which departments. Sounds boring but it's actually useful when you're trying to figure out why the network's crawling and it turns out one team is hogging most of it. It's the same idea as an electricity bill breaking down usage by room, you can actually see where the load is coming from instead of just guessing.

4. Performance Management - A network can technically be "working" and still feel awful to use. This checks for that gap, looking at latency, traffic, bandwidth, whether things are actually running smoothly or just barely holding together when too many people jump on at once. Like a highway that's technically open but jammed at rush hour, still functioning, just not functioning well.

5. Security Management - This is the gatekeeping side of things. Passwords, authentication, who's allowed on the network and who isn't. If some laptop nobody recognizes tries to connect, this is the layer that stops it before it becomes a problem. Basically a bouncer at the door, checking ID before letting anyone in.

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2. Problem & Solution Mapping

1. Fault Management

Problem: Line outages aren't getting caught until someone complains.

The fix here is setting up SNMP based monitoring, something like PRTG or Nagios, with ping checks running on the important switches and routers. The second a device stops responding, an alert goes out automatically instead of everyone finding out the hard way.

2. Configuration Management

Problem: Switch settings keep getting changed with zero record of it.

This is where a tool like SolarWinds NCM comes in, or even just turning on command logging through AAA/TACACS+ on the switches. Either way, every change gets a timestamp and a username attached to it, so if something breaks, you actually know what got touched and by who.

3. Accounting Management

Problem: IT has no clue which department is eating up the network.

NetFlow or sFlow monitoring solves this, paired with something like PRTG or ntopng. These tools break bandwidth usage down by IP range or VLAN, which usually lines up closely with departments since each one tends to sit on its own subnet. This makes it easy to pull up a report and see exactly who's using what.

4. Performance Management

Problem: The network drags during peak hours.

SNMP polling tracks interface utilization, latency, and packet loss over time, tools like PRTG or Zabbix handle this well. Wireshark helps zoom in during peak windows to pinpoint what's actually clogging things up, whether it's one device hogging bandwidth or just too many users at once.

5. Security Management

Problem: Random unknown devices keep joining the network.

802.1X port based authentication or MAC filtering on the switches takes care of this. Basically nothing gets in unless it has the right credentials first, no exceptions. This stops unauthorized laptops or phones from getting network access even if they're just plugged into an open port.

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Task 1.2: Network Design Comparison

Hierarchical	
Ease of Expansion	**High** scalability because it is divided into 3 independent layer which are Core, Distribution, and Access Layer. New network segments can be added without disrupting the existing network.
Reliability	Core and Distribution layers **support redundancy** that allows the rest of the network to stay online. Only the local area of the broken Access switch is affected
Cost & Maintenance	**Moderate to high** cost because specialised hardware needs to be purchased for different layers. However, the maintenance is less strained because of the structured design.

Centralised	
Ease of Expansion	Depending on the main central router has **low to medium** scalability because scaling up is usually done by replacing the central hardware with a more powerful one.
Reliability	**Low** reliability with a single point of failure. If the central hardware breaks the entire network goes down with it.
Cost & Maintenance	**Cheaper** to set up because basic devices can be used but upgrading the central hardware in the future can be very **expensive**.

Distributed	
Ease of Expansion	**High** scalability because independent nodes can be easily added to share the workload but with an increased complexity.
Reliability	**Very high** fault tolerance with no single point of failure. Traffic are automatically rerouted if a router breaks.
Cost & Maintenance	**Expensive** to set up because of complex wiring connections. In addition, it is complex to manage and troubleshoot.

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Scenario: A growing college campus connecting 3 separate buildings to a main server room.

For a multi-building campus, I recommend using the **Hierarchical model** because:

1. Scalable, each building can be treated as an independent entity. New hardware or buildings can be added easily to support current and future infrastructure. For example, each building gets its own **Access** switches for classroom and **Distribution** switches to accumulate the building traffic and that switch is then plugged directly into the **Core** main server room without disrupting each building's own internet connection.

2. Highly reliable from the layered structure that helps contain a network issue only to that specific area. If Building A is having a network problem it will not affect Building B and Building C.

3. Efficient traffic management by dividing networks into specified roles. The **Access** Layer handles local users, **Distribution** layer handles routing between buildings, and **Core** layer serves as the backbone for moving data. Furthermore, it prevents bottlenecks and ensures reliable access to main server room for all students and staff

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Part 2: Advanced SNMP Protocol & Security Investigation

Task 2.1: SNMP Fundamentals

SNMP Polling

Firstly, let's find out more about SNMP Polling. Polling in network management consists of a device asking for and receiving information. In direct contrast to standard SNMP traps, "SNMP polling" is a process that involves the master requesting and receiving data, in the form of Management Information Base (MIB) objects, from the agents deployed at remote sites.

SNMP polling involves retrieving these MIB variables from devices to determine faulty behavior or connection problems. Faulty devices or connections are diagnosed by applying predefined formulas to the extracted MIB variables, making it an essential tool in proactive network management.

So, for example, your SNMP management station could be polling a device, such as a router, to know the status of a network interface. You can configure your manager to perform this task every couple of seconds or minutes (depending on your urgency and network bandwidth). If your interface utilization is at 100%, then your manager will trigger an alarm and send you and your techs alarms in relation to that.

It can also be explained analogically. Whereas a teacher is asking students about their name, their class or personal details. And the student will give their response based on the specific questions asked. In short, SNMP Polling is basically a server asking devices like routers and switches for their current state and information.



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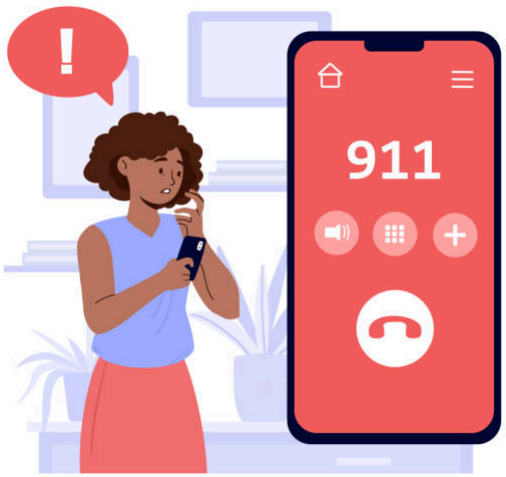
SNMP Traps

An SNMP Trap is an unsolicited message sent from an "agent" to a master station. It provides a means for devices at your sites to alert your master station to events at your sites, even when the master station has not polled the device.

To understand the usefulness of a trap, we must first take a look at the architecture of a network management system. At the top, you'll have your master station, the device to which all of your remote devices report. Down from here, you'll have "agents," devices that report to the master. Now, in a typical network management system, the master station would have to poll agent devices prompting agents to send status information to the master.

Traps don't require a request from the master. They are only sent when something within the agent changes state. That's why they can result in substantial network traffic savings. This can also improve your response time. For example, what if an agent that is configured to send SNMP Traps sets an alarm at a site? It doesn't have to wait to be polled by a master station to send the alarm update. The master station likewise doesn't have to receive frivolous updates about the agent's other unchanged points.

For SNMP Traps, we can analogically explain it as a person (routers/switches) calling 911 [or 999 if we are in Malaysia] for an emergency. And the Emergency Department (Server) will retrieve the emergency report (unsolicited event notification) and write the alert details into a local log file or database for later review.



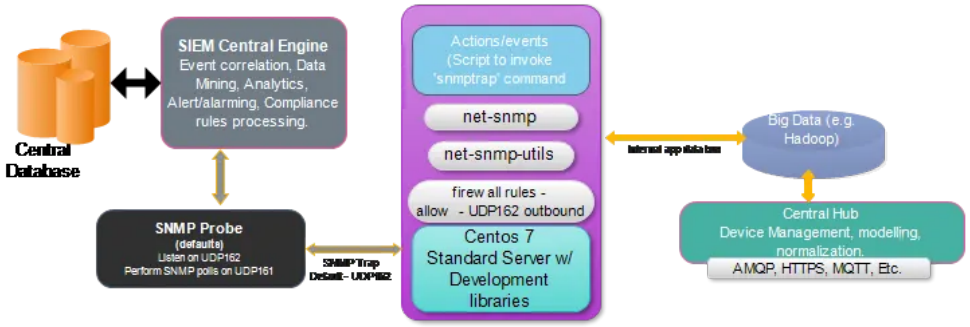
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SNMP Trap Example

To explain how an SNMP Trap works in a practical example, and why it is superior to periodic polling for urgent events. This example analyzes an IoT-SIEM integration scenario based on Dan Tambe's article on medium (2017).

In a Smart City deployment across the City of Dallas, thousands of Internet of Things (IoT) temperature sensors monitor local environmental conditions. Sensor IoT_Temp_Guage_1051 records a temperature of 104°F, breaching its safe threshold by 10°F.

Instead of waiting for the central monitoring server to ask for a status update, the IoT Gateway instantly generates and pushes an unsolicited SNMP Trap directly to a central Security Information and Event Management (SIEM) collector.

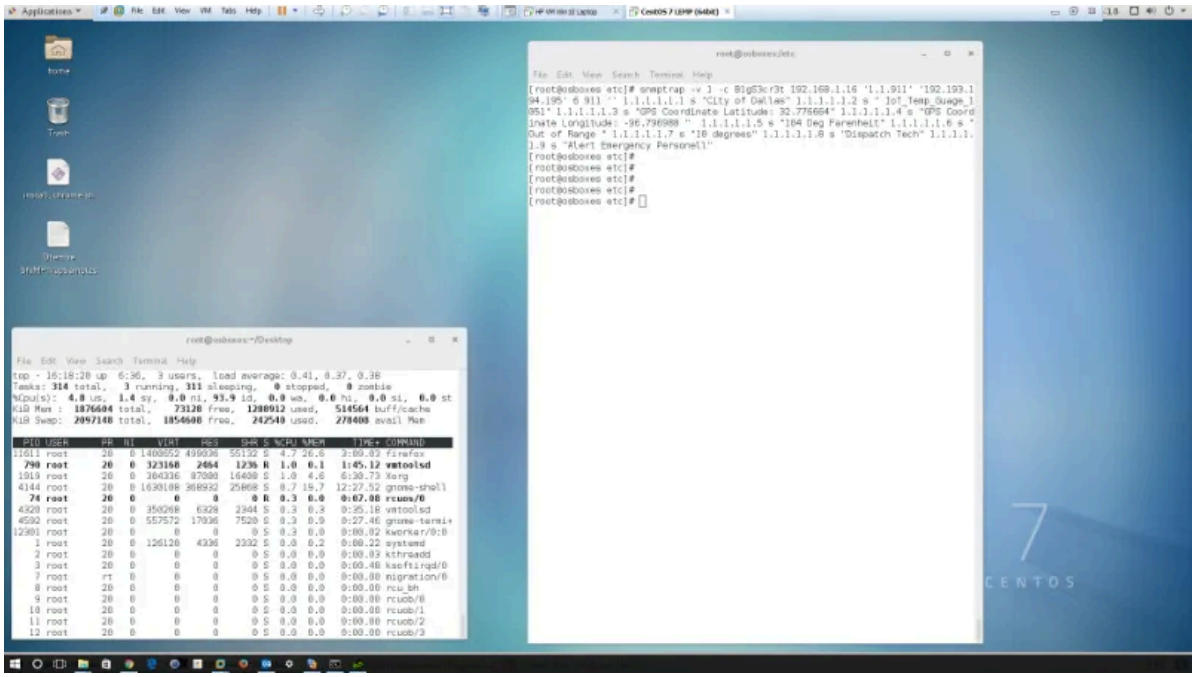


On a CentOS host running the net-snmp utility suite, the gateway executes the following shell command to broadcast the emergency alert:

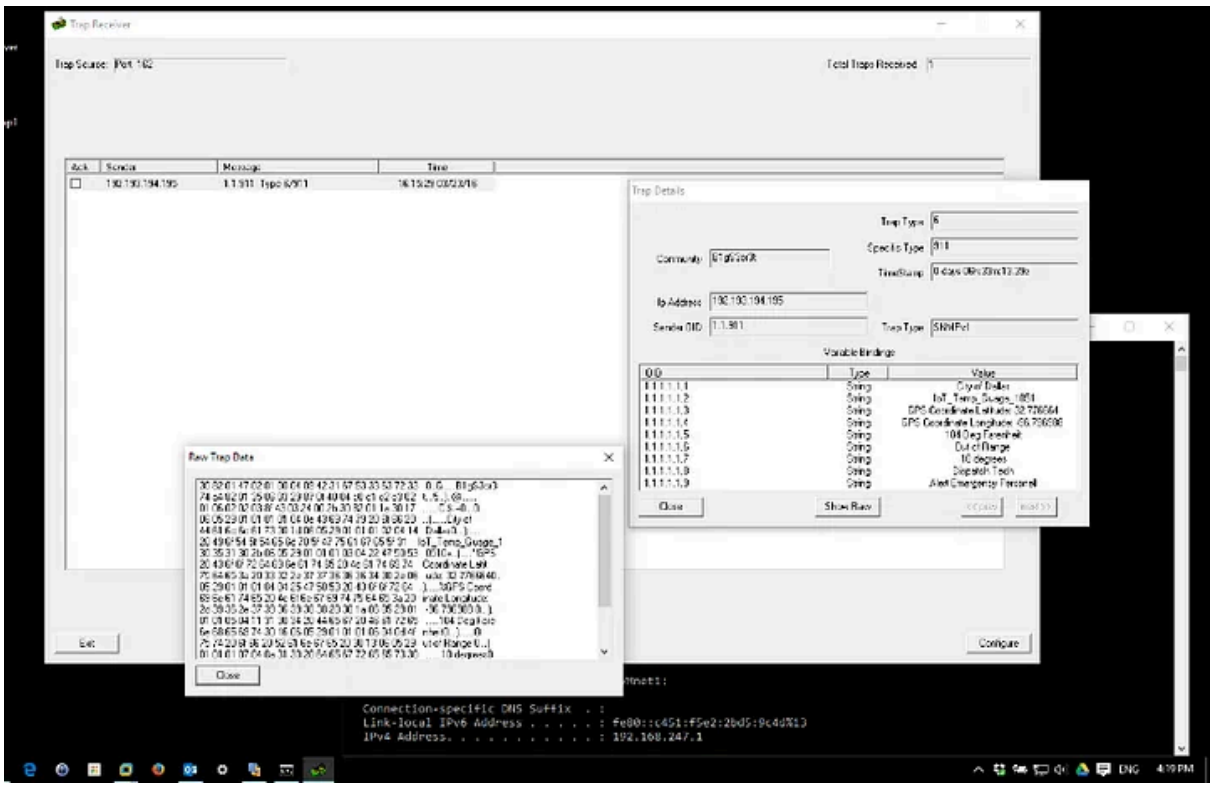
snmptrap -v 1 -c B1gS3cr3t 192.168.1.16 '1.1.911' '192.193.194.195' 6 911 " \ 1.1.1.1.1.1 s "City of Dallas" \ 1.1.1.1.1.2 s "IoT_Temp_Guage_1051" \ 1.1.1.1.1.3 s "GPS Coordinate Latitude: 32.776664" \ 1.1.1.1.1.4 s "GPS Coordinate Longitude: -96.796988" \ 1.1.1.1.1.5 s "104 Deg Fahrenheit" \ 1.1.1.1.1.6 s "Out of Range" \ 1.1.1.1.1.7 s "10 degrees" \ 1.1.1.1.1.8 s "Dispatch Tech" \ 1.1.1.1.1.9 s "Alert Emergency Personnel"

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Proof of Concept:

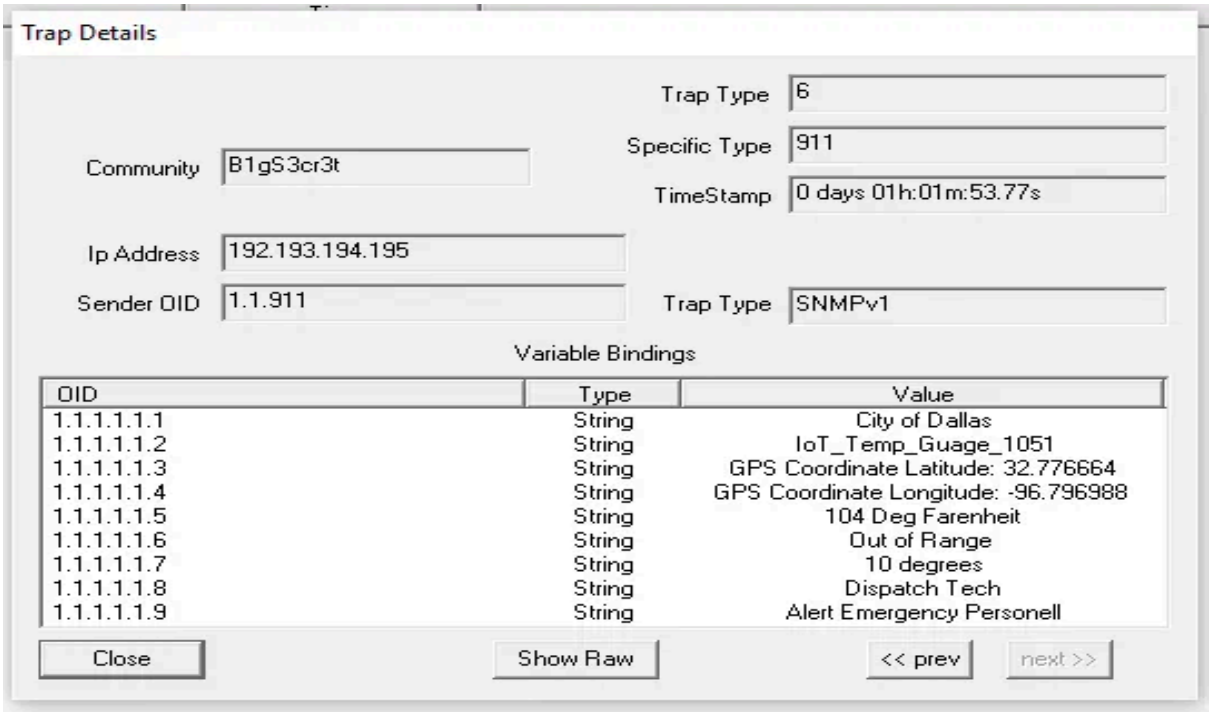


Above shows the command being run in a console and then as the image below shows, it is received immediately as an SNMP Trap on the receiver system. In the example, the receiver is emulating how an SIEM SNMP Probe will function.



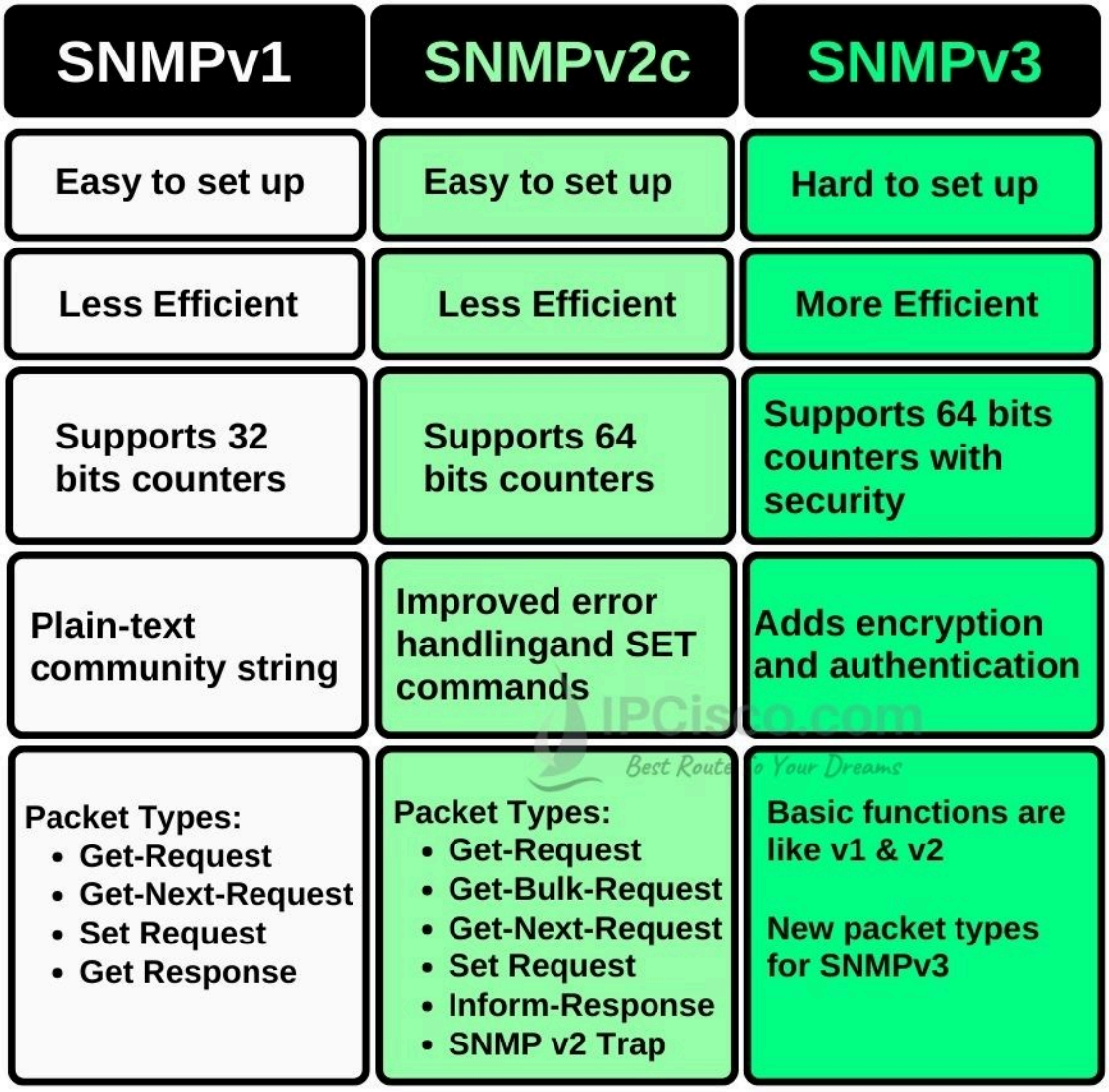
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Once the SNMP trap is received on the SIEM probe side, we can easily decipher each varbind in the SNMP trap as shown in the example below.



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SNMPv3 Advantages



SNMPv1 is the original version of the Simple Network Management Protocol. SNMP was first introduced in 1988 by a group of university researchers to monitor networked devices over TCP/IP networks. By 1990, the Internet Architecture Board (IAB) approved SNMP as an internet standard. In the IT world, SNMPv1 is the equivalent of a dinosaur that hasn't quite gone extinct. It's still in use today as it provides the most basic functionality required for data polling without using significant resources. However, the reason it doesn't require a lot of resources is also a big part of its main drawback. SNMPv1 has very basic security and doesn't include any encryption algorithms.

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SNMPv2 brings 64-bit counters to the table, as well as improvements in security, a greater degree of flexibility when establishing hierarchical management structures, and simpler Management Information Base (MIB) discovery. SNMPv2 also introduces new commands for easier use. This protocol introduced commands like "GETBULK", allowing a manager to request multiple variables from an agent in one message. "INFORM" altered the way that "Traps" worked in SNMPv1. The new "Inform" command requires the manager to respond to the agent, essentially confirming receipt of the message. SSNMPv2 security may be good enough for internal networks, but it should never be considered an option for public or internet-facing devices. It's simple authentication system and lack of encryption makes networks vulnerable to a wide range of threats.

SNMPv3 was developed in 1998 to address mounting security issues. It introduces three new elements: SNMP View, SNMP Groups, and SNMP Users. These three elements ensure that every interaction with a device on the network is effectively authenticated and encrypted, significantly reducing the risk of unauthorized parties gaining access to read/write data. SNMPv3 also introduced encryption methods such as SHA, MD5, and DES to increase security and prevent data tampering and eavesdropping without requiring an overwhelming amount of resources. It should be noted, though, that encryption systems only work if authentication has been enabled.

In terms of security, SNMPv3 is a clear step beyond SNMPv2. Essentially, SNMPv3 is a good option for any network, and should definitely be used on public and internet-facing networks to protect against security risks and threats. SNMPv2 should be limited to low-risk internal networks.

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Task 2.2: Comparative Study of 3 SNMP Implementations

Tool 1: PRTG Network Monitor



1. SNMP Version Supported

SNMP Version Supported	SNMP v1	The simplest and earliest version. It sends data in plain text and utilizes a straightforward community string (password). It is not appropriate for high-load gigabit bandwidth monitoring since it only supports 32-bit counters.
	SNMP v2c	Introduces support for 64-bit counters, allowing accurate bandwidth tracking on high-speed network interfaces. Like v1, it relies on simple community strings without advanced encryption.
	SNMP v3	Provides robust security by adding authentication and encryption capabilities. PRTG supports advanced security options, including SHA-2 (SHA-224, SHA-256, SHA-384, SHA-512) authentication and AES (AES-128, AES-192, AES-256) encryption ciphers.

2. Main Feature

Sensor-Based Architecture	PRTG uses small tools called "sensors" to track specific details. Each sensor focuses on just one job, like checking your computer's memory usage, free hard drive space, or data passing through a single network port.

3. Key Advantage

Ease of Use	PRTG provides a centralized graphical interface for monitoring networks, servers, cloud systems and applications. Its sensor-based design allows administrators to view different monitoring measurements from a single platform.

4. Key Limitation - Performance at Large Scale

Performance at Large Scale	PRTG can require additional resources when monitoring a large number of sensors. SNMPv3 also has performance limitations because encryption increases processing requirements. For larger deployments, monitoring workloads may need to be distributed across multiple probes.

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Tool 2: Net-SNMP

Net-SNMP

1. SNMP Version Supported

SNMP Version Supported	SNMP v1 (RFC 1157)	Legacy protocol support using plain-text community strings for authentication.
	SNMP v2c (RFCs 1901–1908)	Community-based SNMPv2 with improved data counters (64-bit) and bulk retrieval, while retaining plain-text community string security.
	SNMP v3 (RFCs 3411–3418)	Secure framework providing user-based authentication and data encryption such as MD5, SHA authentication, and DES, AES encryption.

2. Main Features

Answering Questions (Polling)	Net-SNMP sits on a device and waits for the administrator's monitoring software to ask for information. When the software asks questions like "How busy is the CPU?" or "How much disk space is left?", Net-SNMP instantly checks the device and sends the answers back.

3. Key Advantage

Flexibility and Customization	Net-SNMP is an open-source implementation that is available for multiple operating systems and provides command-line tools that can be extended or customized for different monitoring requirements.

4. Key Limitation

Command-Line Complexity	Net-SNMP primarily provides command-line tools rather than a built-in graphical monitoring dashboard. Users who require graphs and visual dashboards may need to use additional monitoring software.

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Tool 3: Cisco IOS SNMP



1. SNMP Version Supported

SNMP Version Supported	SNMP v1	Uses basic community strings and plain-text passwords. It offers simple, read-only or read-write access but lacks strong security
	SNMP v2c	Replaces older party-based frameworks with community-based administration. It adds bulk data retrieval and better error handling, though it remains unencrypted.
	SNMP v3	Adds secure user-based authentication and message encryption (privacy). It supports security levels like noAuthNoPriv, authNoPriv, and authPriv

2. Main Features

SNMP Polling	Cisco IOS provides an SNMP agent that allows a network management system to retrieve information from the device. The information is organized through the Management Information Base (MIB), with individual objects identified using Object Identifiers (OIDs). The SNMP manager sends a request for a specific OID, and the Cisco device returns the corresponding value.

3. Key Advantage

Integrated Network Device Monitoring	SNMP functionality is integrated into Cisco IOS network devices, allowing network management systems to monitor device information such as interface statistics, system status and performance without requiring a separate SNMP agent to be installed on the Cisco device.

4. Key Limitation

Cisco-Specific Configuration	Cisco IOS SNMP configuration and management require knowledge of Cisco IOS commands and network-device configuration. Organizations using equipment from multiple vendors may also need additional SNMP management tools to provide centralized monitoring across different platforms.

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